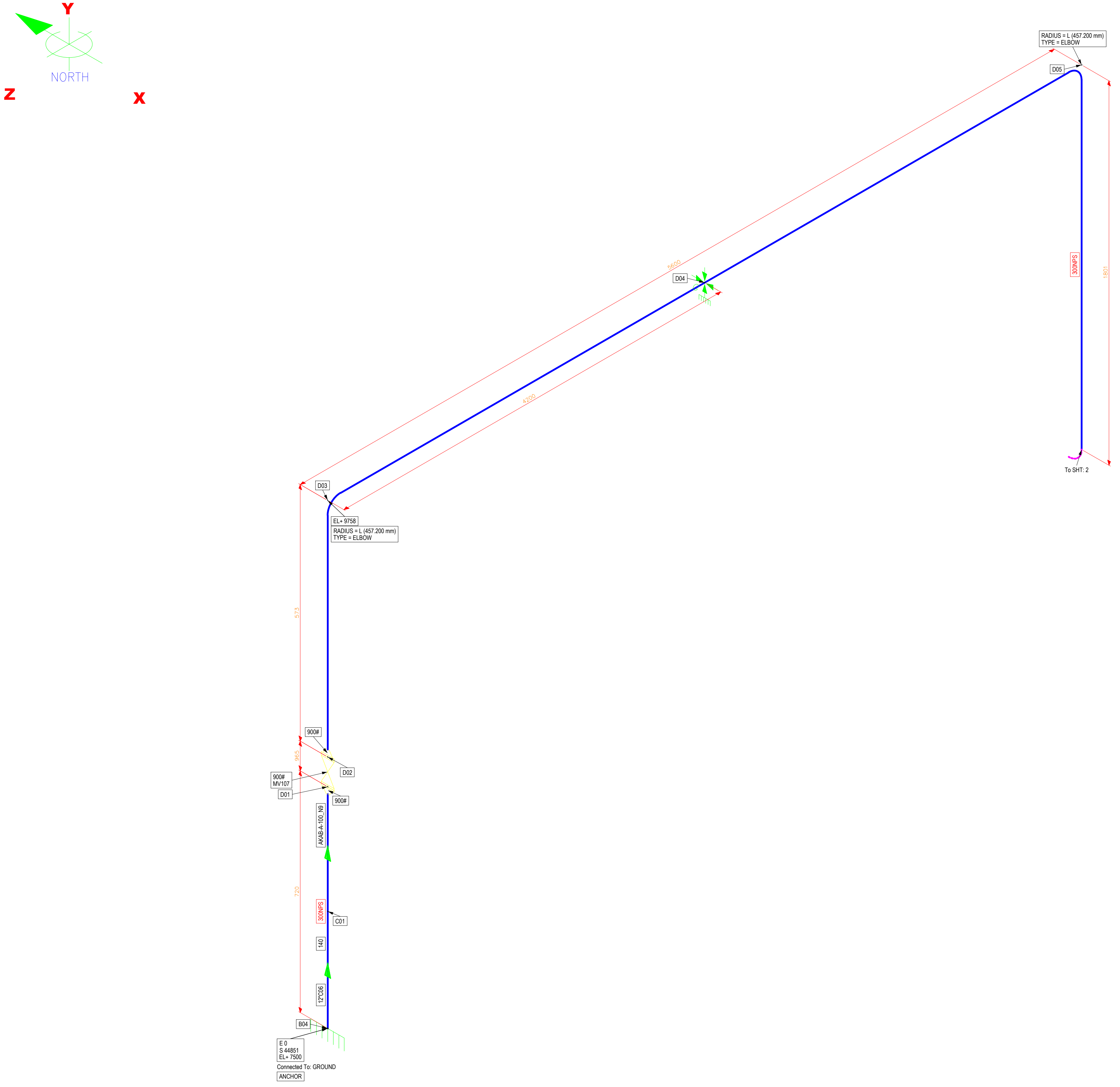


MODEL INFO AutoPIPE Ver: 12.06.00.48
System: AKABEL._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 5 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
12"C06	API5L-B	300.00	140	323.85	28.575	6.000	3.572		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
D01	300.00	CATALOG	900#	1506.86

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
D01	300.00	WELDNECK	900#	168.74	84.37
D02	300.00	WELDNECK	900#	168.74	84.37

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
D04 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
D04 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstp, Guide-V=Up , Lstop, Incl, Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
D04 1	24787.91	15191.23	-2625.67	3.14	3.00	-3.41	Thermal 1{1}
D04 1	-18702.6	-34230.7	4096.61	-2.86		2.62	Thermal 2{1}
D04 1	-18702.6	-36880.6	4096.61	-3.00		2.55	GT2{1}
D04 1	28174.89	18146.99	-3300.36	3.00	3.00	-3.66	GT1P1{1}
D04 1		-2649.92		-0.14		-0.07	Gravity{1}

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
B04	-491.80	-2016.86	7496.68	15699.68	-13297.9	17691.60	Thermal 1{1}
B04	-1040.87	-32015.0	9203.51	14888.93	-12789.4	20622.36	GT1P1{1}

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap	ANCHOR	LINESTOP	V-STOP	ROTATION	GUIDE	INCLINED	SPRING	CONSTANT	DAMPER
Line Direction for Support Gaps									

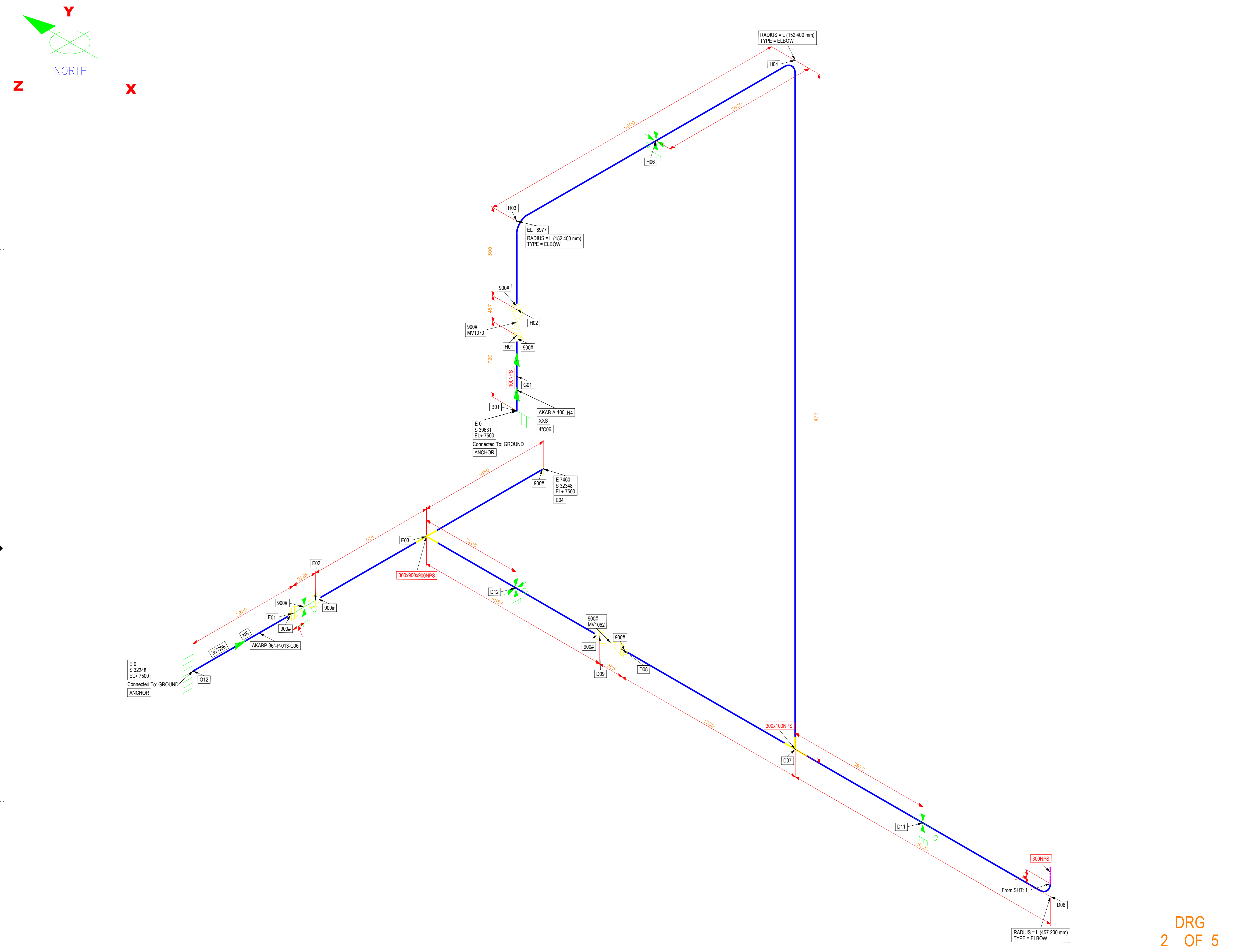
PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I. O	DATE:	
CHECKED BY:	E N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRG. SIZE D

DRAWING No.

REV.



DRG
2 OF 5

MODEL INFO AutoPIPE Ver: 12.06.00.48
System: AKABEL._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 5 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
12"C06	API5L-B	300.00	140	323.85	28.575	6.000	3.572		0.700
36"C06	API5L-X60	900.00	NS	914.40	38.200	6.000	4.775		0.700
4"C06	API5L-B	100.00	XXS	114.30	17.120	6.000	2.140		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
D08	300.00	CATALOG	900#	1506.86
E01	900.00	CATALOG	900#	4838.10
H01	100.00	CATALOG	900#	144.70

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
D08	300.00	WELDNECK	900#	168.74	
D09	300.00	WELDNECK	900#	168.74	84.37
E01	900.00	WELDNECK	900#	1764.00	
E02	900.00	WELDNECK	900#	1764.00	
E04	900.00	BLIND	900#	1200.00	600.00
H01	100.00	WELDNECK	900#	23.13	
H02	100.00	WELDNECK	900#	23.13	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
D11 1	V-stop	Ground	Rigid		2540.00	0.10
D12 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
D12 1	Guide-V	Ground	Rigid		3.00	0.10
E01 M1	V-stop	Ground	Rigid		2540.00	
H06 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
H06 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstp, Guide-V=Up , Lstop, Incl, Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
D07	H-Branch	114.30	17.12	API5L-B					
D07	D-Header	323.85	28.57	API5L-B	Welding			1.199	1.265

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
D11 1		23741.13		-6.09	0.86	4.63	Thermal 2{1}
D11 1	3303.11	-37868.4	-1822.17	9.62		-5.64	GT1P1{1}
D11 1	5206.90	-4129.51	-1088.01	0.56		-0.15	Pressure 3{2}
D11 1		-23741.1				-0.33	Gravity{1}
D12 1	4197.27	-44269.1	-18985.4	2.25		-3.00	GT1{1}
D12 1	4782.16	-44194.7	-23907.0	2.35		-3.00	GT1P1{1}
D12 1		-36390.0				-0.01	Gravity{1}
E01 M1		-164955		-0.45		-2.50	GT1P1{1}
E01 M1		-190692		-0.06		-0.43	GT3P3{2}
E01 M1		-170382					Gravity{1}
H06 1	45.66	-535.34	-33.06	2.94		-2.27	GT1{1}
H06 1	81.33	-400.29	-27.40	3.00		-2.25	GT1P1{1}
H06 1	-169.01	-2580.77	186.64	-1.60		1.61	GT2P2{1}
H06 1	-170.09	-0.10	0.81	-0.18			Wind 2{1}
H06 1		-2024.82		-0.02		-0.13	Gravity{1}

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

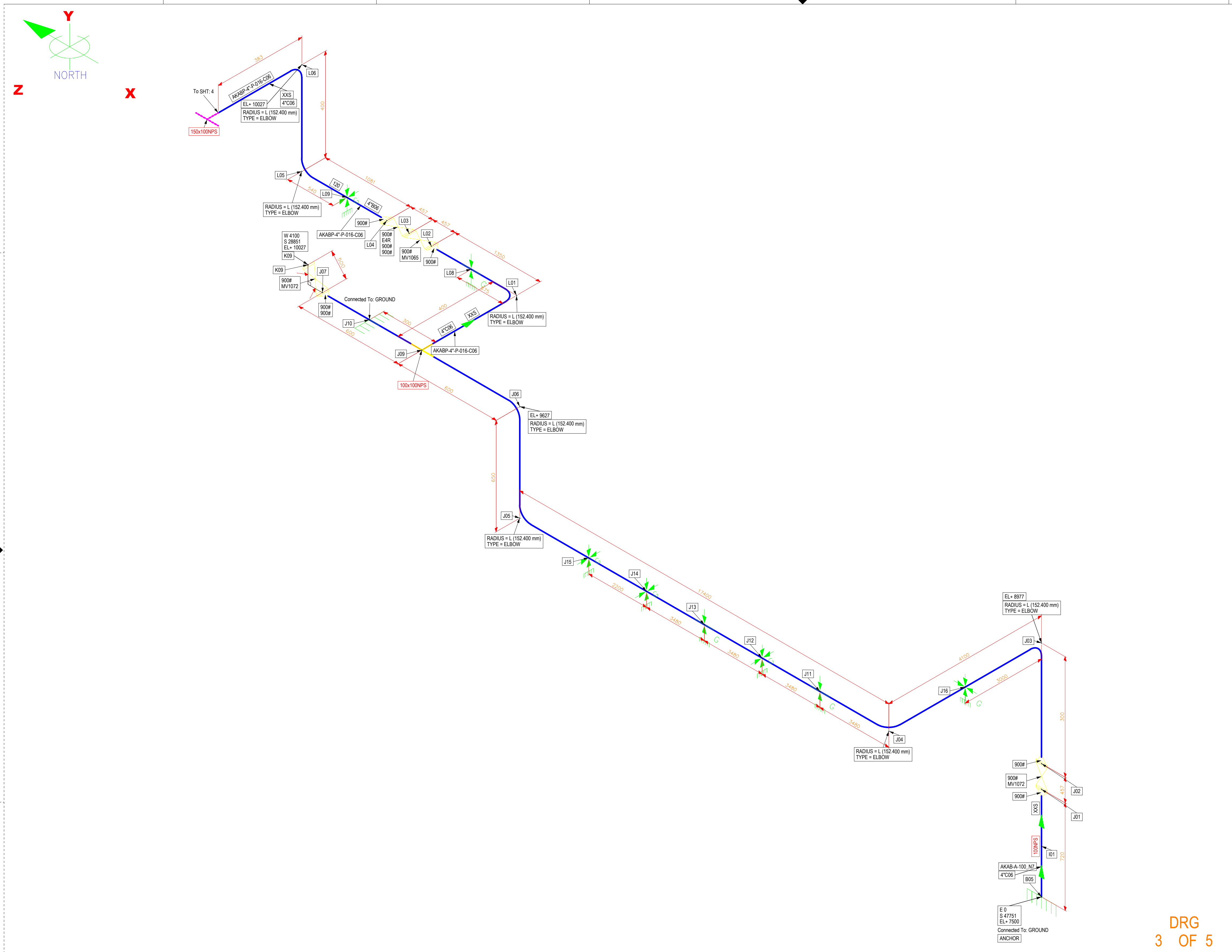
Point	Fx	Fy	Fz	Mx	My	Mz	Combination
B01	196.00	-776.96	1564.78	1111.38	-576.02	-262.12	Thermal 1{1}
B01	190.19	-3362.85	1472.71	1002.78	-582.81	-243.51	GT1P1{1}
B01	197.34	-3297.17	1266.48	808.56	-574.10	-257.16	GT1{1}
O12	-31591.7	12590.23	14269.60	19082.43	149595.0	11037.06	GT1{1}
O12	-4953.90	13747.75	3033.27	22026.57	20860.54	7291.44	GT3P3{2}
O12	20361.90	15311.59	-1050.24	21656.08	-115009	6247.31	GT2{1}
O12	-35490.8	12632.08	18380.67	19122.02	162619.2	10995.56	GT1P1{1}

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap	ANCHOR	LINESTOP	V-STOP	ROTATION	GUIDE	INCLINED	SPRING	CONSTANT	DAMPER
Line Direction For Support Gaps									

PROJECT NO:	AKABEL	DATE:
ANALYZED BY:		DATE:
CHECKED BY:		DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley
DRAWING No.
DRG. SIZE D
REV.



DRG
3 OF 5

MODEL INFO AutoPIPE Ver: 12.06.00.48
System: AKABEL._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 5 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
4"B06	API5L-B	100.00	120	114.30	11.125	6.000	1.391		0.700
4"C06	API5L-B	100.00	XXS	114.30	17.120	6.000	2.140		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
J01	100.00	CATALOG	900#	144.70
J07	100.00	CATALOG	900#	144.70
L02	100.00	CATALOG	900#	144.70
L03	100.00	CATALOG	900#	181.44

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
J01	100.00	WELDNECK	900#	23.13	
J02	100.00	WELDNECK	900#	23.13	
J07	100.00	WELDNECK	900#	23.13	
L02	100.00	WELDNECK	900#	23.13	
L04	100.00	WELDNECK	900#	23.13	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
J11 1	V-stop	Ground	Rigid		2540.00	0.10
J12 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J12 1	Guide-V	Ground	Rigid		3.00	0.10
J13 1	V-stop	Ground	Rigid		2540.00	0.10
J14 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J14 1	Guide-V	Ground	Rigid		3.00	0.10
J15 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J15 1	Guide-V	Ground	Rigid		3.00	0.10
J16 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J16 1	Guide-V	Ground	Rigid		3.00	0.10
L08 1	V-stop	Ground	Rigid		2540.00	0.10
L09 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
L09 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
J09	L-Branch	114.30	17.12	API5L-B					
J09	J-Header	114.30	17.12	API5L-B	Welding			1.000	1.000

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
J11 1	-1.96	-1.96	-173.36	-0.02		-1.47	Wind 4(1)
J11 1	132.15	-2066.10	156.26	12.17		14.42	GT1P1(1)
J11 1	351.56	-7.71	78.08	0.35		0.08	Pressure 3(2)
J11 1		-1731.74		-0.02			Gravity(1)
J12 1	237.38	173.97	1272.65	8.87		3.03	Thermal 1(1)
J12 1	241.62	-1201.50	1315.16	9.05		3.00	GT1P1(1)
J12 1	-139.58	-1460.67	4.49	-0.10		-0.02	GT3(2)
J12 1	283.77	6.92	-6.30	0.25			Pressure 3(2)
J12 1		-1376.10		-0.02		-0.03	Gravity(1)
J13 1	0.67	2.11	155.16	0.02		3.62	Wind 3(1)
J13 1	164.25	-1771.35	-63.25	5.94		-2.35	GT1P1(1)
J13 1	-0.67	-2.07	-155.58	-0.02		-3.55	Wind 4(1)
J13 1	-182.15	-1824.62	21.29	-3.58		0.36	GT2(1)
J13 1	314.41	-14.29	-3.06	0.14			Pressure 3(2)
J13 1		-1553.77		-0.02		-0.06	Gravity(1)
J14 1	-0.81	-1236.52	-123.65	-0.03		-2.32	GW4(1)
J14 1	-2.59	14.03	1.27	0.06	0.02	-0.07	Pressure 1(1)
J14 1				2.82	0.02	-1.50	GT1P1(1)
J14 1	0.82	-10.49	125.74	0.02		2.30	Wind 3(1)
J14 1	-26.30	-1358.36	-0.24	-0.02		-0.07	GT3(2)
J14 1		-1246.86		-0.02		-0.07	Gravity(1)
J14 1	148.45	87.33	0.25	0.04			Pressure 3(2)
J15 1	-1.57	-911.70	-91.16	-0.03		-0.90	GW4(1)
J15 1		840.59		-0.22	1.38	0.34	Thermal 2(1)
J15 1	717.40	-7180.77	-208.60	0.85		-0.25	Thermal 1(1)
J15 1	742.11	-8207.43	-223.68	0.85		-0.33	GT1P1(1)
J15 1		-840.59		-0.02		-0.07	Gravity(1)
J16 1	18513.85	832.85	1744.99	3.02		1.54	Thermal 1(1)
J16 1	-9626.05	-2368.64	-991.66	-3.00		-1.16	GT2(1)
J16 1	19092.30	-1053.90	1777.21	3.00		1.60	GT1P1(1)
J16 1		-1906.72		-0.02		0.02	Gravity(1)
L08 1	-166.39	-2786.72	-223.39	-0.29		-0.37	GT1P1(1)
L08 1	307.78	-3823.46	214.16	0.27		0.20	GT2(1)
L08 1	311.75	-3791.23	214.65	0.26		0.19	GT2P2(1)
L08 1		-2921.74		-0.01			Gravity(1)
L09 1	-367.44	-545.73	23.84	-1.91		0.12	Thermal 1(1)
L09 1	-360.96	-3628.51	21.65	-1.94		0.09	GT1P1(1)
L09 1	-68.77	-128.41	-277.06	-0.01		-0.04	Wind 4(1)
L09 1	-367.44	-3668.07	23.84	-1.92		0.10	GT1(1)
L09 1	-371.92	-31.45	-48.31	-0.03			Pressure 3(2)
L09 1		-3122.34		-0.01		-0.03	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and
movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
B05	-3770.00	-3295.28	-3023.66	-2830.54	-1178.89	5094.85	GT1P1(1)
J10	2404.97	-2874.65	1828.89	470.72	653.12	2277.21	GT2(1)

G-Gap
Line Direction For Support Gaps

ANCHOR	LINESTOP	V-STOP	ROTATION	GUIDE	INCLINED	SPRING	CONSTANT	DAMPER

REV.	DATE	REVISION	BY	CHKD	APP

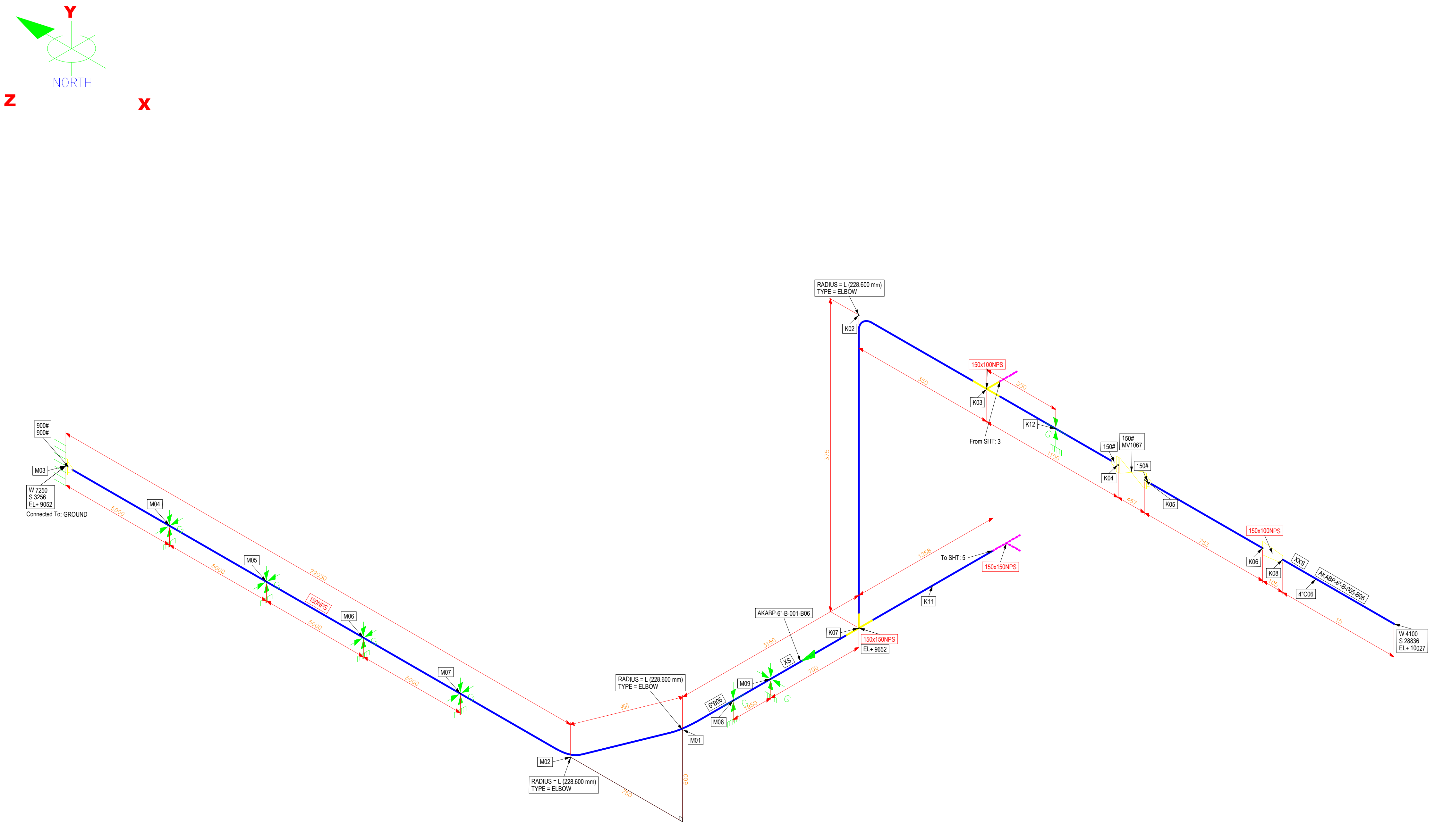
PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I. O	DATE:	
CHECKED BY:	E N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRG. SIZE D

DRAWING No.

REV.



MODEL INFO AutoPIPE Ver: 12.06.00.48
System: AKABEL._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 5 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
4"C06	API5L-B	100.00	XXS	114.30	17.120	6.000	2.140		0.700
6"B06	API5L-B	150.00	XS	168.28	10.973	6.000	1.372		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
K04	150.00	CATALOG	150#	153.32

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
K04	150.00	WELDNECK	150#	11.34	5.67
K05	150.00	WELDNECK	150#	11.34	5.67
M03	150.00	WELDNECK	900#	49.90	24.95

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
K12 1	V-stop	Ground	Rigid		2540.00	0.10
M04 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
M04 1	Guide-V	Ground	Rigid		3.00	0.10
M05 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
M05 1	Guide-V	Ground	Rigid		3.00	0.10
M06 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
M06 1	Guide-V	Ground	Rigid		3.00	0.10
M07 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
M07 1	Guide-V	Ground	Rigid		3.00	0.10
M08 1	V-stop	Ground	Rigid		2540.00	0.10
M09 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
M09 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstp, Guide-V=Up , Lstop, Incl, Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
K03	L-Branch	114.30	17.12	API5L-B					
K03	K-Header	168.27	10.97	API5L-B	Welding			1.430	1.574
K07	M-Header	168.27	10.97	API5L-B	Welding			1.430	1.574

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
K12 1		2635.93		-1.35	0.10	0.26	Thermal 1{1}
K12 1				-1.41	0.11	0.25	GT1P1{1}
K12 1	3.78	-73.88	241.62			0.06	Wind 3{1}
K12 1	630.21	-6296.22	13.68	1.38		0.03	GT2{1}
K12 1	630.21	-3660.29	13.68	1.42		0.03	Thermal 2{1}
K12 1		-2635.93		-0.04			Gravity{1}
M04 1	-230.20	-2511.10	13.37	-2.96		0.17	GT2{1}
M04 1	0.18	-1.06	249.15			1.72	Wind 3{1}
M04 1	-290.08	-2912.91	-0.37	-0.05			GT3{2}
M04 1	580.70	1.41	0.44	0.08			Pressure 3{2}
M04 1		-2490.96					Gravity{1}
M05 1	-214.11	-2400.00	13.07	-5.92		0.35	GT2{1}
M05 1	3.04	3.76	546.54			3.01	Wind 3{1}
M05 1	288.64	-2890.75	-0.19	0.05		-0.01	GT3P3{2}
M05 1	576.18	-4.98	-2.29	0.16			Pressure 3{2}
M05 1		-2471.02				-0.01	Gravity{1}
M06 1	-273.32	-2810.25	-8.74	-8.88		-0.30	GT2{1}
M06 1	3.17	-14.04	455.12			3.02	Wind 3{1}
M06 1	-297.65	-2985.98	-1.66	-0.16		-0.03	GT3{2}
M06 1	594.01	18.64	-0.68	0.24			Pressure 3{2}
M06 1		-2544.65				-0.02	Gravity{1}
M07 1	-335.26	1521.64	-2464.47	-11.84		-2.96	Thermal 2{1}
M07 1	443.04	-986.48	1528.43	8.24		3.04	Thermal 1{1}
M07 1	-335.26	-701.76	-2464.47	-11.84		-3.00	GT2{1}
M07 1	455.07	-3241.80	1596.71	8.47		3.00	GT1P1{1}
M07 1	507.98	-118.04	91.73	0.33		0.06	Pressure 3{2}
M07 1		-2223.40				-0.04	Gravity{1}
M08 1	56.86	-10.90	-97.33	0.04		-0.07	Wind 4{1}
M08 1	-192.51	-1999.98	-44.57	-11.20		-2.61	GT2{1}
M08 1	277.75	129.97	46.14	0.28		0.05	Pressure 3{2}
M08 1		-1204.98		-0.03		-0.02	Gravity{1}
M09 1		800.08		-1.13	0.40	-1.43	Thermal 2{1}
M09 1				-1.16	0.40	-1.45	GT2{1}
M09 1	56.22	-1102.93	183.12	0.48		1.55	Thermal 1{1}
M09 1	57.53	-1920.73	177.91	0.48		1.55	GT1P1{1}
M09 1		-800.08		-0.03		-0.02	Gravity{1}

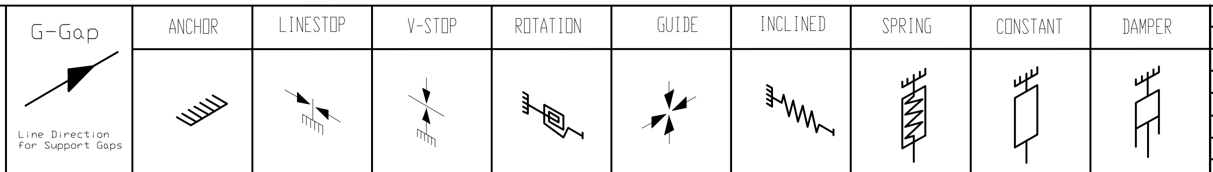
For Rotational Supports: Loads are moments (ft-lb) and
movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
M03	-288.75	-2710.47	-466.62	19.81	1040.34	-1034.67	GW4{1}
M03	-261.77	-0.26	-466.66	14.60	1040.01	-0.43	Wind 4{1}
M03	-1337.91	-2920.35	-0.08	7.85	1.20	-1209.32	GT3P3{2}
M03	5771.35	5.01	10.05	-106.64	-63.76	8.31	Thermal 2{1}

REV.	DATE	REVISION	BY	CHKD	APP



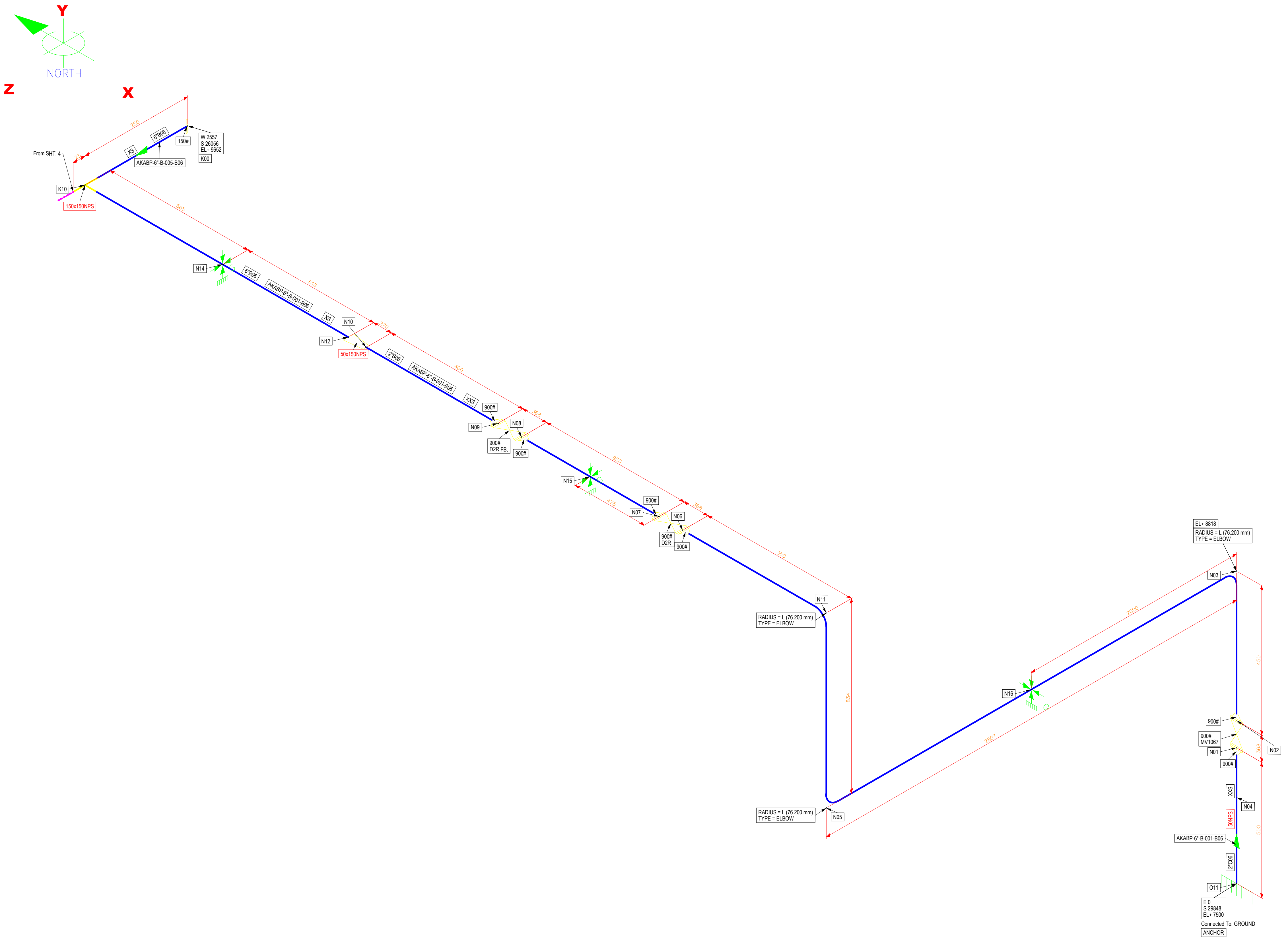
PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I. O	DATE:
CHECKED BY:	E N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRG. SIZE D

DRAWING No.

REV.



MODEL INFO AutoPIPE Ver: 12.06.00.48
System: AKABEL._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 5 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
2"B06	API5L-B	50.00	XXS	60.325	11.074	6.000	1.384		0.700
2"C06	API5L-B	50.00	XXS	60.325	11.074	6.000	1.384		0.700
6"B06	API5L-B	150.00	XS	168.28	10.973	6.000	1.372		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
N01	50.00	CATALOG	900#	48.08
N06	50.00	CATALOG	900#	95.26
N08	50.00	CATALOG	900#	48.08

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
K00	150.00	BLIND	150#	12.70	
N01	50.00	WELDNECK	900#	10.90	5.45
N02	50.00	WELDNECK	900#	10.90	5.45
N06	50.00	WELDNECK	900#	10.90	5.45
N07	50.00	WELDNECK	900#	10.90	5.45
N08	50.00	WELDNECK	900#	10.90	5.45
N09	50.00	WELDNECK	900#	10.90	5.45

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
N14 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
N14 1	Guide-V	Ground	Rigid		3.00	0.10
N15 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
N15 1	Guide-V	Ground	Rigid		3.00	0.10
N16 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
N16 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
K10	N-Branch	168.27	10.97	API5L-B					
K10	K-Header	168.27	10.97	API5L-B	Welding			1.430	1.574

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
N14 1		1211.15		5.19	0.03	1.21	Thermal 2{1}
N14 1				5.25	0.03	1.24	GT2{1}
N14 1	4.93	-10.75	109.38			0.08	Wind 3{1}
N14 1	-162.37	-1634.35	-15.69	-4.34		-0.40	GT1P1{1}
N14 1		-1211.15		0.06		0.02	Gravity{1}
N15 1	209.77	-2245.88	260.39	4.03		3.00	GT2{1}
N15 1	209.77	-515.42	260.39	3.98		2.45	Thermal 2{1}
N15 1		-1730.46		0.06		0.55	Gravity{1}
N16 1	-45.35	23.88	77.35	-0.61		1.05	Thermal 1{1}
N16 1	1.04	-942.15	4.37	-0.54		-0.34	GT3P3{2}
N16 1	83.71	18.71	-33.74	1.40		-0.56	Thermal 2{1}
N16 1		-921.15		-0.54		-0.34	Gravity{1}

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Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
O11	69.36	-876.52	-32.37	-70.34	-17.80	-82.61	GW1{1}
O11	-36.34	99.18	261.65	202.39	26.58	29.74	Thermal 2{1}
O11	35.88	-1013.12	-365.41	-319.12	-25.28	-57.48	GT1P1{1}
O11	-75.58	-875.21	-33.42	-71.83	-17.85	36.77	GW2{1}

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REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction For Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:		DATE:	
CHECKED BY:		DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRAWING No.

DRG. SIZE D

REV.